Replicating BERTSCORE, and Finding Where It Stops Working

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Abstract

BERTSCORE (Zhang et al., 2020) evaluates generated text by matching contextual embeddings rather than exact n-grams, and claims stronger correlation with human judgement than surface-overlap metrics. I reproduce that claim on the original WMT18 data, where BERTSCORE outperforms BLEU in all seven to-English language pairs, and where the mean margin between the two metrics matches the published figure. I then test the claim on two datasets the authors did not use. On SummEval the metric generalises to summarisation but tracks content coverage more than twice as well as factual correctness. On a perturbation corpus I construct from arXiv abstracts, a sentence whose meaning has been inverted by an antonym substitution scores at or above its meaning-preserving counterpart 42.4% of the time, while merely shuffling word order costs 0.65. Adding a small instruction-tuned language model as an LLM judge reduces that inversion failure to 12.7%, but the judge scores scrambled word salad at 0.93. The two methods have complementary blind spots: BERTSCORE is sensitive to surface form and largely blind to meaning reversal, and the LLM judge is the reverse. Code and data: https://github.com/jeniljasani9-netizen/appln-of-ds-assignment1

1 The Paper and Why It Was Chosen

1.1 The Problem It Addresses

Systems that generate text must be evaluated somehow, and asking humans to read every output does not scale. The standard substitute is an automatic evaluation metric: a function taking a machine-produced *candidate* and one or more human-written *references*, and returning a number intended to stand in for a human judgement.

For two decades the dominant metrics have measured *surface overlap*. BLEU (Papineni et al., 2002) counts shared n-grams; ROUGE (Lin, 2004) does the same with recall in place of precision; METEOR (Banerjee and Lavie, 2005) adds stemming and a small synonym lexicon. All ultimately ask whether the same word forms appear in both sentences, which produces two well-known failure modes. A candidate can be an excellent paraphrase yet score near zero because it uses different words, and a candidate can be a poor output yet score well because reordering leaves unigrams intact.

1.2 The Method

BERTSCORE keeps the shape of these metrics, computing a precision, a recall and an F-measure by matching candidate tokens to reference tokens, but replaces *exact* matching with *soft similarity* between contextual embeddings.

A contextual embedding model such as BERT (Devlin et al., 2019) or RoBERTa (Liu et al., 2019) maps each token to a vector whose value depends on its surrounding words. Given a reference $x = \langle x_1, \ldots, x_k \rangle$ and candidate $\hat{x} = \langle \hat{x}_1, \ldots, \hat{x}_l \rangle$, both passed through the same model, BERTSCORE computes all pairwise cosine similarities and applies *greedy matching*: each token is paired with its single most similar counterpart in the other sentence, independently of every other pairing:

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} x_i^\top \hat{x}_j \tag{1}$$

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} x_i^\top \hat{x}_j \tag{2}$$

$$F_{\text{BERT}} = 2 \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}} \tag{3}$$

Embeddings are pre-normalised to unit length, so $x_i^\top \hat{x}_j$ is the cosine similarity. Two refinements are offered: inverse document frequency weighting, which discounts function words, and baseline rescaling, a strictly monotonic linear transforma-

tion that improves score readability without changing any ranking.

The greedy matching step is cheap and permissive about word order, but it is not a bijection, and nothing in it checks whether a matched pair agrees or contradicts. Section 5 shows that this property is directly visible in how the metric behaves.

1.3 Justification

The paper appeared at the International Conference on Learning Representations in 2020. ICLR is rated **A*** in the CORE/ICORE rankings, the top band, awarded to roughly 8% of ranked venues in the CORE2023 round. Google Scholar Metrics corroborates this independently: ICLR is ranked **second** among all venues in the Artificial Intelligence subcategory of Engineering & Computer Science, with an h5-index of 362 and an h5-median of 652 (checked 18 August 2026). It trails only NeurIPS on h5-index and exceeds it on h5-median.

Google Scholar reports 12,202 citations for the paper as at 18 August 2026. Beyond raw counts, the metric has become a standard baseline in the literature that followed it: MOVERSCORE (Zhao et al., 2019), BLEURT (Sellam et al., 2020) and COMET (Rei et al., 2020) all benchmark against it.

The reference implementation is at `https://github.com/Tiiiger/bert_score` under the MIT licence, with approximately 1,900 stars and 240 forks. It is distributed on PyPI as `bert-score` (version 0.3.13) and integrated into the Hugging Face `evaluate` library. Critically for a replication project, BERTSCORE requires only forward passes, with no training, so the computational cost is within reach of a student machine. The human judgement data it was validated on is also public, which means the original result is checkable rather than merely reported.

2 Datasets Used in the Original Paper

2.1 Why Metric Evaluation Needs Judged Data

A point the paper assumes rather than states is that evaluating a *metric* is a different activity from evaluating a *system*. There is no ground-truth correct score for a sentence. Instead the metric is treated as a predictor of human judgement, and its quality is the *correlation* between its scores and human scores over the same outputs. The resources required are therefore not ordinary text corpora but corpora of *system outputs paired with human ratings*. This constraint shapes every dataset decision in this project.

`lp`	`cs-en`
`mt`	The monument, a dignified piecemeal place, stands by the will of history practically in th…
`ref`	The memorial, a solemn place of commemoration, stands there thanks to the will of history,…
`raw`	53.0
`score`	-0.1455

Figure 1: One row of the WMT18 segment-level data. `mt` is the machine translation, `ref` the human reference, `raw` the Direct Assessment score given by human judges, and `score` its z-normalised form. Text truncated for display.

2.2 WMT18 Metrics Task

The primary evaluation corpus is the WMT18 metrics task (Ma et al., 2018), containing the outputs of 149 translation systems across 14 language pairs: English to and from Czech, German, Estonian, Finnish, Russian, Turkish and Chinese. Each candidate comes with a gold reference and human judgements at two granularities. Segment-level judgements score individual candidate–reference pairs; system-level judgements assign one score per system. The human judgements derive from Direct Assessment, in which annotators rate translations on a continuous scale rather than ranking them.

Figure 1 shows one row of this data as used in the replication.

2.3 Additional Original Datasets

Three further resources appear in the paper. WMT16 (Bojar et al., 2016) and WMT17 add 202 systems reported in the appendix; WMT16 also serves as a *validation* set for selecting which embedding layer to extract from, which matters for replication because the widely used default (`roberta-large`, layer 17) is a tuned hyperparameter rather than an arbitrary choice. The COCO 2015 Captioning Challenge supplies twelve systems captioning the COCO validation set (Lin et al., 2014), correlated against two system-level human measures following Cui et al. (2018). Finally, QQP and PAWS-QQP (Zhang et al., 2019) provide an adversarial paraphrase task in which non-paraphrase pairs share almost all vocabulary while differing in meaning.

2.4 Correlation Measures

Following WMT convention the paper uses absolute Pearson correlation at the system level and Kendall's rank correlation τ at the segment level. Pearson measures linear agreement, appropriate where data points are few and human scores are continuous. Kendall's τ measures agreement in ordering:

$$\tau = \frac{|\text{Concordant}| - |\text{Discordant}|}{|\text{Concordant}| + |\text{Discordant}|} \qquad (4)$$

where a pair of segments is concordant if the metric orders them as the human did. Because competing metrics often differ by small margins, the paper tests significance with the Williams test at the system level and bootstrap resampling at the segment level, following Graham and Baldwin (2014).

3 Replication

3.1 Environment

All experiments ran in a GitHub Codespaces container: 2 CPU cores, 8 GB RAM, Ubuntu Linux, Python 3.14. The package was installed with `pip install bert-score` (PyPI version 0.3.13) against PyTorch 2.14 and Transformers 5.17 (Wolf et al., 2020).[1] The repository pins exact versions in `requirements.txt`. No GPU was used at any point.

A reproducibility observation worth recording: the repository's last commit predates this project by roughly two years, which would ordinarily raise the prospect of dependency friction. In practice the package installed and ran without modification and with no version pinning required. A research artefact that still works unmodified on libraries released two years after its last commit is unusually robust.

Two smaller observations. The library reports its own version as `0.3.12` in its configuration hash although PyPI installed `0.3.13`, so the internal version string appears not to have been updated at release. This matters because that hash is the standard way of identifying a scoring configuration in published work. Separately, loading `roberta-large` emits a report that the pooler weights are absent and newly initialised; this is benign, because BERTSCORE extracts hidden states from layer 17 and never uses the pooler, but it could easily be misread as a failed load.

Scoring throughout used the default English configuration with baseline rescaling, reported by the library as `roberta-large_L17_no-idf_version=0.3.12 (hug_trans=5.17.0)-rescaled`, confirming that the tuned layer-17 setting discussed in Section 2 is the shipped default.

3.2 Obtaining the Data

The dataset link published on the WMT18 metrics task page for segment-level Direct Assessment data, hosted on a personal academic page at Dublin City University, returns HTTP 404 and appears to have been retired. This is itself a reproducibility finding: the data underpinning a 2020 result is no longer available at the address the original shared task advertises. The judgements were instead obtained from a maintained mirror on the Hugging Face Hub (`RicardoRei/wmt-da-human-evaluation`), filtered to year 2018.

Resource constraints limited the replication to a subset. I sampled 300 candidate–reference pairs for each of the seven to-English language pairs, 2,100 in total. The paper used between roughly 5,000 and 78,000 pairs per language pair, so this is a substantially smaller sample, and Section 7 returns to what that costs.

3.3 Results

Table 1 gives the replication result. BERTSCORE outperforms BLEU in all seven language pairs, on both Pearson and Kendall measures, with no exceptions. Overall segment-level Kendall τ is 0.369 against 0.238, a relative improvement of roughly 55% in rank agreement with human judges.

3.4 Comparison with the Published Figures

Table 2 places these numbers beside the paper's Table 4, to-English column. Individual language pairs differ, in some cases substantially: et-en and tr-en are higher here than published, cs-en lower.

Two honest reasons account for the per-pair variation. First, sample size: 300 pairs per language gives a standard error on τ of roughly ± 0.05, which explains some but not all of the spread. Second, and more importantly, the estimators differ. The paper computes segment-level agreement using daRR, a relative-ranking statistic derived from DA judgements by forming pairs whose scores differ by a

[1] All code, data and captured outputs: https://github.com/jeniljasani9-netizen/appln-of-ds-assignment1

	Pearson r		Kendall τ	
Pair	BERT	BLEU	BERT	BLEU
cs-en	0.461	0.244	0.294	0.139
de-en	0.610	0.470	0.454	0.349
et-en	0.659	0.392	0.490	0.290
fi-en	0.380	0.248	0.265	0.163
ru-en	0.487	0.336	0.325	0.241
tr-en	0.638	0.409	0.432	0.268
zh-en	0.405	0.301	0.311	0.203
All	**0.521**	0.348	**0.369**	0.238

Table 1: Correlation with WMT18 segment-level human judgements, 300 pairs per language pair. BERTSCORE (F_{BERT}) exceeds BLEU in every cell.

	F_{BERT}		BLEU	
Pair	Paper	Mine	Paper	Mine
cs-en	0.404	0.294	0.233	0.139
de-en	0.550	0.454	0.415	0.349
et-en	0.397	0.490	0.285	0.290
fi-en	0.296	0.265	0.154	0.163
ru-en	0.353	0.325	0.228	0.241
tr-en	0.292	0.432	0.145	0.268
zh-en	0.264	0.311	0.178	0.203
Mean gap	**+0.131**		**+0.131**	

Table 2: Segment-level Kendall correlations against the published values (Zhang et al., 2020, Table 4, to-English). The bottom row is the mean margin between the two metrics: the quantity the paper's claim is actually about.

threshold, whereas I compute standard Kendall τ directly against the continuous z-normalised scores. These are different quantities and their values are not interchangeable.

What does replicate, and what the paper's claim is actually about, is the *margin* between the two metrics. Averaged across the seven pairs, BERTSCORE's advantage over BLEU is $+0.131$ in my run and $+0.131$ in the published table. The agreement to three decimal places is partly fortuitous given the estimator mismatch, but the direction and approximate magnitude of the finding reproduce cleanly.

3.5 Investigating the Original Result

Beyond the aggregate, it is worth examining where the metric succeeds and fails. Scoring the reference *"the weather is cold today"* against the candidate *"it is freezing today"* gives $P = 0.787$, $R = 0.668$, $F_1 = 0.727$. The pair shares no bigram, so BLEU scores it at zero. This is the paper's motivating case reproduced exactly: the metric recognises that *freezing* and *cold* are semantically close.

The converse case is more revealing. Scoring the reference *"the treatment reduced mortality in the trial"*:

Candidate	F_1
the treatment reduced mortality in the trial	1.0000
the **therapy** reduced mortality in the trial	0.9571
the treatment **increased** mortality in the trial	0.9392
the treatment **did not reduce** mortality…	0.8553
trial the in mortality reduced treatment the	0.5059
the stock market closed higher on Friday	0.1464

Table 3: Scores against a single reference. A meaning-preserving synonym substitution and a meaning-inverting antonym substitution differ by 0.018, while scrambling the identical words costs 0.494.

A faithful paraphrase and a statement asserting the opposite clinical result are separated by 0.018. Destroying the grammar while preserving every word costs 0.494. This observation motivated the constructed dataset in Section 4.

4 Construction of New Data

Two new datasets were used, asking different questions. One is an existing resource the paper did not use, testing whether the claim generalises to another task. The other I constructed, testing the mechanism behind the claim.

4.1 SummEval

SummEval (Fabbri et al., 2021) pairs 16 summarisation systems on 100 CNN/DailyMail articles with expert human ratings on four *separately scored* dimensions: coherence, consistency, fluency and relevance. Because the dimensions are rated independently, it can answer a question WMT18 cannot, namely which aspect of quality BERTSCORE actually tracks. My project proposal hypothesised that the metric would correlate better with relevance than with factual consistency, on the grounds that a summary can closely resemble a reference while stating a fact incorrectly.

I scored 30 articles $\times$ 16 systems, 480 summaries, each against three human reference summaries, taking the maximum score across references as the paper does for multi-reference settings. The dataset is obtained at runtime from the Hugging Face Hub (`mteb/summeval`) under its published licence.

4.2 The arXiv Perturbation Corpus

The second dataset is constructed, and its design choice is deliberate: by generating the damage my-

sev	type	candidate
0	identical	We evaluate coding agent under a safety constraint, where each task pairs a ma…
1	synonym	We evaluate **code** agent under a safety constraint, where each task pairs a mani…
2	scramble	each agent task an with the evaluate robot under a safety pairs coding not a W…
5	negation	**It is not the case that** we evaluate coding agent under a safety constraint, wh…

Figure 2: Variants generated from one source sentence. This sentence received no antonym or number-swap variant: it contains no digits, and no WordNet antonym was found for any of its taggable content words. Text truncated for display.

self I know the relative severity of each variant by construction, so *no human annotation is required*. Without funding for crowdsourced labels this is what makes a meaning-sensitivity experiment feasible at all. The trade-off is that the severity ordering is a design assumption rather than a measured human judgement, which Section 5 shows the data itself challenges.

Construction proceeds in three scripted stages. First, `scrape_arxiv.py` retrieves 100 recent abstracts from each of three arXiv categories (`cs.CL`, `q-bio.NC`, `econ.EM`) through the public arXiv API, respecting the documented three-second rate limit. Scientific abstracts are a deliberate domain shift: BERTSCORE was validated on news translation and image captions, and technical vocabulary is where contextual embeddings might plausibly be weakest.

Second, abstracts are split into sentences and filtered to those between 10 and 30 words, ending in a full stop, free of LaTeX markup, inline citations, URLs and unbalanced parentheses. This yields 1,731 clean candidates, of which 200 are retained.

Third, `build_perturbations.py` applies six transformations to each sentence. Severity 0 is an unchanged copy; severity 1 substitutes a content word for a WordNet synonym (Miller, 1995); severity 2 shuffles word order while preserving the exact word multiset, in the spirit of PAWS (Zhang et al., 2019); severity 3 alters a numeric value; severity 4 substitutes a WordNet antonym; severity 5 negates the claim, either by inserting *not* after an auxiliary or by prepending an explicit denial. This produces 1,008 variants.

Two transformations have incomplete coverage, and this is a real limitation rather than a presentational detail. Number swaps applied to only 24 of 200 sentences, because most abstract sentences contain no digits. Antonym substitution applied to 184, because WordNet antonym coverage is uneven across parts of speech.

	Pearson r		Kendall τ	
Dimension	BERT	BLEU	BERT	BLEU
Relevance	**0.358**	0.205	0.239	0.165
Coherence	0.347	0.137	0.221	0.086
Fluency	0.164	0.079	0.106	0.034
Consistency	**0.150**	0.101	0.111	0.062

Table 4: SummEval correlations, 480 machine summaries. Relevance and consistency are bolded for contrast: the metric predicts content coverage more than twice as well as factual correctness.

4.3 Inter-Annotator Agreement

The marking criteria ask for an agreement measure where human annotation was used. No human annotation was used in constructing either dataset, so no agreement statistic applies. For the perturbation corpus the severity ordering is established by construction, as described above. For SummEval the human ratings are those released by Fabbri et al. (2021), collected from three expert annotators per summary; I use their published scores and did not collect new judgements.

5 Results on New Data

5.1 SummEval: Coverage, Not Correctness

Table 4 reports correlation with each of the four expert-rated dimensions. BERTSCORE beats BLEU on all four, so the paper's central claim does generalise from translation to summarisation, a task the authors never tested.

The more interesting result is the spread across dimensions. Relevance reaches $r = 0.358$ and consistency only 0.150, a difference of $+0.208$. Consistency in SummEval means factual alignment with the source article. The metric therefore predicts whether a summary *covers the right content* more than twice as well as whether it *states that content correctly*. This confirms the hypothesis recorded in the project proposal, and it is the same weakness the constructed corpus isolates below, arriving from a completely different direction.

5.2 The Perturbation Corpus

Table 5 gives mean scores by perturbation type. The pattern does not follow the severity ordering I designed.

Three findings stand out.

sev	Perturbation	n	BERTSCORE	BLEU
0	identical	200	1.000	100.0
1	synonym	200	0.933	87.7
3	number swap	24	0.972	87.9
4	antonym	184	0.923	87.5
5	negation	200	0.880	78.6
2	scramble	200	0.283	9.6

Table 5: Mean scores on the perturbation corpus, 1,008 variants, ordered by BERTSCORE rather than by designed severity. Meaning-altering edits cluster near the meaning-preserving baseline; only surface disruption separates.

Perturbation	n	BERTSCORE	LLM judge
identical	60	1.000	0.988
synonym	60	0.928	0.945
scramble	60	**0.296**	**0.934**
number swap	8	0.967	0.503
antonym	55	**0.915**	**0.609**
negation	60	0.876	0.337
Spearman w/ severity		-0.502	$\mathbf{-0.770}$
Antonym $\geq$ synonym		47.3%	**12.7%**

Table 6: The two methods on the same 303 pairs. Bolded rows show where each fails: BERTSCORE on scrambling, the judge on meaning inversion. Their failures are complementary.

Meaning inversion is barely penalised. A synonym swap, which preserves meaning, scores 0.933. An antonym swap, which inverts it, scores 0.923. The difference is 0.010. Comparing each sentence against its own counterpart, the antonym variant scores *at or above* the synonym variant in 78 of 184 cases, **42.4%**: close to a coin flip on whether an inverted sentence looks better than a faithful one. Negation costs only 0.052. Changing a number scores 0.040 *higher* than a synonym swap, though with $n = 24$ this is suggestive rather than conclusive; a plausible mechanism is that numerals occupy a tight region of embedding space, so one numeral substitutes for another more cleanly than any genuine synonym pair does.

Surface disruption is penalised enormously. Scrambling word order, which preserves the exact word multiset and destroys only syntax, collapses the score to 0.283, a drop of 0.650 from the synonym baseline. The metric is thus roughly sixty times more sensitive to word order than to meaning reversal. This traces directly to the greedy matching step described in Section 1: each token selects its best partner independently, and nothing in that procedure examines whether the pairing preserves or reverses the proposition.

The advantage over BLEU does not survive. Spearman correlation with designed severity is -0.502 for BERTSCORE and -0.508 for BLEU. On this dataset the two are equivalent, and BLEU is fractionally ahead. Excluding the scramble group, both rise to around -0.75, with BLEU still marginally higher. This is a negative result for the paper's claim in a specific and bounded sense: the advantage demonstrated on human-judged translation quality does not transfer to detecting controlled meaning damage.

I should also record that my own severity scale is partly refuted by these numbers. I assigned scrambling severity 2 and negation severity 5, assuming shuffled word order damages meaning less than reversing a claim. Both metrics disagree violently. The scale conflates two distinct kinds of damage, surface disruption and semantic inversion, and separating them is what makes the finding visible.

5.3 LLM-as-Judge

The final evaluation introduces a method unavailable when the paper was written: prompting a language model to act as judge (Zheng et al., 2023; Liu et al., 2023). Here it serves as a *competing metric*, tested on exactly the data where BERTSCORE was shown to fail.

The judge is Qwen2.5-0.5B-Instruct (Qwen Team, 2025), prompted with the reference, the candidate, and the question *"Does the candidate preserve the meaning of the reference? Answer Yes or No."* Rather than sampling text, I read the model's next-token distribution and compute $P(\text{Yes})/(P(\text{Yes}) + P(\text{No}))$, summing over capitalisation and leading-space token variants. This makes the score continuous and fully deterministic, and costs one forward pass per item. 303 pairs from 60 source sentences were judged in 489 seconds.

Two configuration decisions were forced by the hardware and are worth recording for anyone reproducing this. Loading the model in bfloat16, the obvious choice for memory, ran roughly forty times slower than float32, because PyTorch lacks optimised bf16 CPU kernels. Qwen2.5-1.5B exhausts 8 GB of RAM in float32, so the 0.5B model is the largest that fits. Published LLM-as-judge work uses GPT-4 class models, so this result should be read as a floor rather than a ceiling.

Table 6 shows the comparison. On the test BERTSCORE fails, the judge is markedly better:

the antonym variant scores at or above its synonym counterpart in 7 of 55 cases (12.7%) against 26 of 55 (47.3%), close to a four-fold reduction. Spearman with severity improves from -0.502 to -0.770. Negation drops to 0.337, indicating the judge detects reversal directly.

But the scramble row inverts the picture. BERTSCORE collapses to 0.296 on scrambled text; the judge scores it 0.934, barely distinguishing word salad from a faithful paraphrase. Reading the prompt explains why: the judge is asked whether meaning is preserved, and a scrambled sentence retains every content word, so a small model recovers the gist and answers yes.

The conclusion is therefore not that one method supersedes the other. BERTSCORE is sensitive to surface form and largely blind to meaning reversal; the LLM judge is sensitive to meaning and largely blind to surface disruption. Their failure modes are complementary, which suggests combining them rather than replacing one with the other, since neither alone covers both.

6 Ethical Considerations

This project involved no human participants, no personal information and no identifiable data, so it required no human research ethics approval. Several other considerations apply, and I organise them around the Australian Code for the Responsible Conduct of Research (National Health and Medical Research Council et al., 2018), a principles-based document setting out eight principles (P1–P8) of responsible research conduct together with 29 responsibilities, of which R14–R29 apply to researchers rather than to institutions. The principles most relevant here are P1 honesty, P2 rigour, P3 transparency, P4 fairness in the treatment of others, and P7 accountability.

6.1 Issues Arising in the Original Paper

The BERTScore paper carries no ethics statement. This is unsurprising for ICLR 2020, since the practice of requiring one became widespread only afterwards (Benotti and Blackburn, 2022), and it is an observation about publishing norms rather than a criticism of the authors. Two issues are nonetheless visible in retrospect.

The first concerns the human judgements the paper depends on. WMT Direct Assessment scores and COCO captioning judgements were collected from crowdworkers whose compensation, consent and working conditions are not discussed in the paper, because the paper consumes datasets gathered by others. Responsibility R26 of the Code asks researchers to acknowledge those who have contributed to the research, and P4 concerns fairness in the treatment of others. A metric paper rests entirely on annotation labour that is invisible in its results tables, and the convention of citing the dataset rather than the annotators makes that labour easy to overlook. The same observation applies to the present report, which cites three such datasets.

The second concerns what the metric is for. An evaluation metric is infrastructure: it is used to select models, to tune systems, and to report progress. A metric adopted as widely as this one, with over 12,000 citations, shapes which systems the field decides are good. Its failure modes therefore propagate into every downstream decision made using it, which is a different and larger consequence than an error in a single model.

6.2 Issues Arising in This Work

Dual use. My perturbation corpus is, viewed one way, a documented recipe for generating text that defeats a widely deployed metric. Responsibility R23 asks that findings be disseminated responsibly, accurately and broadly, which in a disclosure case requires weighing the two halves against each other. I judge publication justified on the same grounds as Li et al. (2024), who disclosed a vulnerability in transformer gradient sharing: the failure mode exists whether or not it is documented, and the field cannot develop defences against a weakness it has not been shown. The construction uses only WordNet substitutions and word reordering, techniques already standard in the adversarial NLP literature (Zhang et al., 2019), so the marginal capability released is small relative to the diagnostic value.

Consequences of the finding itself. This is the consideration I weigh most heavily. Table 3 shows the metric assigning 0.939 to *"the treatment increased mortality"* against a reference stating it reduced mortality, barely below the 0.957 given to a faithful paraphrase. If BERTSCORE is used to select or tune summarisation systems in medical, legal or news settings, factual inversions of exactly this kind would go largely unpenalised while the system optimises for surface similarity. The honest framing is that this is a property of similarity-based metrics generally, not a defect specific to one paper, and the SummEval consistency result suggests the

same conclusion from real human ratings.

Data provenance. Three sources were used. WMT18 judgements are released publicly by the shared task organisers for exactly this purpose. SummEval is released under its authors' published licence. arXiv abstracts were retrieved through the official API, respecting its documented rate limit, and abstracts are distributed by arXiv for programmatic access. No personal data was collected at any stage: arXiv author names were never extracted, only abstract text. Under P3 transparency and R27, which asks that other relevant work be cited and acknowledged accurately, all three sources are cited here and the retrieval scripts are in the public repository. R22 asks that clear, accurate and complete records of research data be retained; every generated dataset and every captured program output is committed to that repository rather than described second-hand.

Honest reporting. Two results in this report are unfavourable to the method being studied and to my own design. BLEU slightly outperforms BERTSCORE on my own corpus (Section 5), and my severity scale is partly refuted by the data. Both are reported rather than omitted. P1 honesty and P2 rigour both extend explicitly to the *reporting* of research, not only to its conduct, and R21 asks that conclusions be justified by the results. The latter is why the number-swap row, with $n = 24$, is described as suggestive rather than treated as a finding. In a replication study specifically, selective reporting would defeat the purpose of the exercise.

Use of generative AI. Consistent with the unit's policy and with P7 accountability, which covers the development, undertaking and reporting of research alike, I used an AI assistant during this project for code drafting, debugging and editorial feedback on this report. All experiments were executed by me, every numerical result reported here was produced by the scripts in the repository and checked against their captured outputs, and the cited references were verified against their sources. Responsibility for the content rests with me.

7 Limitations

Sample size. The WMT18 replication uses 300 pairs per language pair where the original used thousands, giving a standard error on Kendall τ of roughly ± 0.05. SummEval uses 30 of 100 articles. The LLM judge evaluation covers 303 of 1,008 variants. All three were constrained by CPU-only compute; the SummEval run alone took 62 minutes.

Estimator mismatch. As noted in Section 3, the paper reports daRR and this report reports Kendall τ on z-normalised DA scores. The per-pair values are therefore not directly comparable to the published table, and I have compared margins rather than absolute values accordingly.

Perturbation coverage. The number-swap transformation reached only 24 of 200 sentences, falling to 8 in the LLM judge subset. No conclusion should be drawn from that row. Antonym substitution reached 184 of 200, limited by WordNet coverage.

Judge model size. Qwen2.5-0.5B is at the very bottom of the range used in LLM-as-judge work. Its strong performance on meaning inversion is therefore a lower bound, and its weak performance on scrambled text may partly reflect model capacity rather than a property of the approach. Only one prompt was tested, and LLM judges are known to be prompt-sensitive.

Single domain for constructed data. The perturbation corpus draws from three arXiv categories but only from scientific abstracts. Whether the same pattern holds for conversational or narrative text is untested.

8 Conclusions and Reflections

The paper's central claim replicates. On its own data, BERTSCORE agrees with human judges better than BLEU in all seven to-English language pairs, and the mean margin between the two metrics matches the published figure. The reference implementation, unmodified since 2024, installs and runs against current libraries without pinning, which is a better reproducibility outcome than this project anticipated.

What the new data adds is a boundary on that claim. Three independent lines of evidence converge on the same weakness. SummEval shows the metric tracking content coverage more than twice as well as factual correctness. The perturbation corpus shows a meaning-inverted sentence outscoring its faithful counterpart 42.4% of the time, while shuffled word order is penalised sixty times more heavily. And the LLM judge, which does detect inversion, fails in exactly the opposite way. The

mechanism is visible in the method itself: greedy matching pairs each token with its nearest neighbour and never asks whether the pairing agrees or contradicts.

Two reflections on doing the work. First, the most informative results came from the places where my own design was wrong. My severity scale assumed scrambling damages meaning less than negation, and the data refuted it; that refutation is what made the surface-versus-meaning distinction visible at all. Second, the constraint that shaped this project most was not compute but *judged data*. Evaluating a metric requires system outputs paired with human ratings, which cannot be scraped, and the obsolescence of the original WMT18 download link is a reminder of how fragile that infrastructure is. Designing a dataset whose ground truth comes from construction rather than annotation was the decision that made the central experiment possible without funding.

The practical conclusion is that BERTSCORE measures whether the right words are present, not whether what they say is true. For ranking translation systems, which is what it was validated for, that is adequate. For any setting where factual inversion matters, it is not sufficient alone, and the complementary failure modes documented here suggest that combining an embedding metric with a model-based judge would cover ground that neither covers by itself.

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